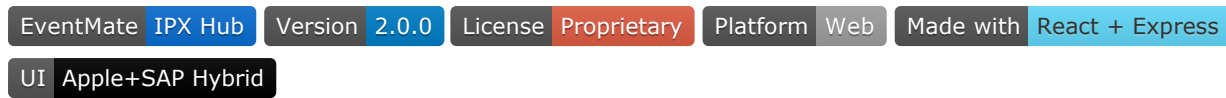


EventMate — Intelligent Participant Experience (IPX) Hub

A complete hackathon solution that enhances the participant experience at SAP events through intelligent registration, personalized session recommendations, AI-powered assistance, and seamless check-in.



EventMate is an intelligent event management platform designed specifically for SAP events. It provides a seamless end-to-end experience for attendees, from registration to check-in and session management, with AI-powered personalized recommendations and intelligent assistance.

Key Features

🌟 Overview

EventMate is a comprehensive event management platform designed specifically for SAP events and hackathons. It provides a seamless end-to-end experience for attendees, from registration to check-in and session management, with AI-powered personalized recommendations and intelligent assistance.

![EventMate Banner](./docs/banner.png)

Technical Stack

- **Frontend:** React (Create React App), React Router, Tailwind CSS
- **Backend:** Express.js

🚀 Key Features- **Database:** File-based mock DB (**db.json**)

- **Authentication:** HMAC-SHA256 signature for QR codes
- **Dynamic Registration System:** Multi-step form with real-time validation and personalization options- **Utilities:** Calendar integration (ICS, Google Calendar, Outlook)
- **SIT Concierge Chatbot:** AI-powered chatbot using Google's Gemini LLM with rule-based fallback- **Testing:** Jest for API endpoints
- **Personalized Agenda:** Smart recommendations based on attendee interests and behavior
- **Secure QR Code Check-in:** HMAC-signed QR codes for secure venue access### Screenshots
- **Networking Assistant:** Connect attendees with similar interests and backgrounds

- **Calendar Integration:** Export sessions to iCalendar, Google Calendar, or Outlook | Landing Page | Registration | Dashboard |
- **Admin Dashboard:** Comprehensive management tools for event organizers | :---😞:---😞:---😞
- **Mobile-Responsive Design:** Optimized experience across all devices | 🖼️Landing | 🖼️Registration | 🖼️Dashboard |

🔧 Technology Stack | Session Detail | Check-in | Admin View |

|:---😞:---😞:---😞

Frontend | 🖼️Session | 🖼️Checkin | 🖼️Admin |

- **Framework:** React.js
- **Routing:** React Router v6## Architecture
- **Styling:** Tailwind CSS with custom SAP UI Theme
- **Animation:** Framer Motion🖼️Architecture Diagram
- **State Management:** React Context API & Hooks
- **Form Handling:** React Hook FormThe application follows a client-server architecture:
- **Testing:** Jest & React Testing Library

1. **Frontend (React):** User interface with components and pages

Backend2. **Backend (Express.js):** RESTful API endpoints

- **Server:** Express.js3. **Mock Database:** JSON file-based data store
- **Authentication:** Custom HMAC implementation, Firebase Auth support4. **Service Layer:** Business logic, recommendation engine, QR code generation
- **AI Integration:** Google Gemini API
- **Data Storage:** JSON file-based (with ready migration to SAP HANA)## Getting Started
- **API Documentation:** OpenAPI/Swagger
- **Testing:** Jest#### Prerequisites

📱 Application Screens- Node.js (v14+)

- npm or yarn

Installation

```

<tr>

  <td align="center"><strong>Landing Page</strong></td>1. Clone the
repository:

  <td align="center"><strong>Registration</strong></td>

  <td align="center"><strong>Dashboard</strong></td>```bash
</tr>git clone https://github.com/yourusername/eventmate.git

<tr>cd eventmate

  <td></td>```

  <td></td>

  <td></td>2. Install backend dependencies:

</tr>

<tr>```bash

  <td align="center"><strong>SIT Concierge</strong></td>cd backend

  <td align="center"><strong>Check-in Portal</strong></td>npm install

  <td align="center"><strong>Admin View</strong></td>```

</tr>

<tr>3. Install frontend dependencies:

  <td></td>

  <td></td>```bash

  <td></td>cd ../frontend

</tr>npm install

```

...

Running the Application

Architecture

1. Start the backend server:



```
Architecture Diagram```bash
```

```
cd backend
```

```
npm start
```

The application follows a modern client-server architecture:```\n

1. **Frontend Layer:** React components with responsive UI/UX design
2. Start the frontend development server:
2. **API Gateway:** RESTful endpoints with proper error handling
3. **Business Logic Layer:** Core event management functionality```\n
4. **AI Services:** Recommendation engine and Gemini-powered chatbotcd frontend
5. **Data Layer:** Structured storage with JSON (expandable to databases)npm start

```
## 🚦 Getting Started
```

```
3. Open your browser and navigate to `http://localhost:3000`
```

```
### Prerequisites
```

```
## Usage Guide
```

```
- Node.js (v14.0.0 or newer)
```

```
- npm/yarn### Demo Credentials
```

```
- Google Gemini API Key (optional for enhanced chatbot)
```

```
- Staff Check-in Portal: Access code `SAP2025`
```

```
### Installation- Admin Dashboard: Access code `ADMIN2025`
```

```
#### Backend Setup### Registration Flow
```

```
```bash1. Complete the registration form
```

```
Clone the repository2. Receive QR code for check-in

git clone https://github.com/yourusername/eventmate.git3. Add event to
calendar

cd eventmate

Check-in Process

Install backend dependencies

cd backend1. Staff scans attendee's QR code

npm install2. System verifies HMAC signature

3. Attendee is marked as checked in

Configure environment

cp .env.example .env### Admin Features

Edit .env with your configuration

- View all attendees and their check-in status

Start backend server- Export attendee data as CSV

npm run dev- View event statistics
```

## Security Features

### Frontend Setup

- QR codes signed with HMAC-SHA256

```
Navigate to frontend directory- Session storage for admin credentials

cd ../frontend

Future Enhancements

Install frontend dependencies

npm install- **Real AI Integration**: Replace rule-based recommendations
with ML model

- **Mobile App**: Native mobile experience with push notifications
```

```
Start development server- Offline Support: Enable check-in without internet connection
```

```
npm start- Advanced Analytics: Detailed insights for event organizers
```

## Contribution

### Environment Configuration

This project was created as a hackathon prototype. Feel free to fork and extend!

Create a `.env` file in the backend directory with:

## License

```
PORT=5000This project is licensed under the MIT License – see the LICENSE file for details.
```

```
HMAC_SECRET=your_secure_hmac_secret_for_qr_code_signing
```

```
GEMINI_API_KEY=your_gemini_api_key_here---
```

```
NODE_ENV=development
```

```
```Created with ♥ for the SAP Hackathon
```

```
## 🔧 API Integration
```

```
### SIT Concierge Chatbot
```

The intelligent chatbot uses Google's Gemini API for natural language understanding and contextual responses:

```
```javascript
// Example API usage
const response = await fetch('/api/chat', {
 method: 'POST',
 headers: {
 'Content-Type': 'application/json'
 },
 body: JSON.stringify({
 message: "When is the next keynote?",
 attendeeId: "att_12345"
 })
});
```

## Personalized Recommendations

```
// Get personalized session recommendations
const recommendations = await
fetch(`/api/attendee/${attendeeId}/recommendations`);
```

## Testing

```
Run backend tests
cd backend
npm test

Run frontend component tests
cd frontend
npm run test:components

Run frontend integration tests
npm run test:integration
```

## Security Features

- **HMAC Signatures:** All QR codes are cryptographically signed to prevent tampering
- **Environment Variables:** Sensitive information stored in environment, not in code
- **Input Validation:** All user inputs validated server-side
- **Security Headers:** Proper HTTP headers to prevent common web vulnerabilities

## Use Cases

### 1. Event Registration & Check-in

- Streamlined registration process
- Secure and quick check-in with QR codes

### 2. Event Navigation & Information

- AI-powered chatbot assistant
- Personalized session recommendations

### 3. Networking & Community Building

- Connect attendees with similar interests
- Facilitate meaningful professional connections

## Roadmap

- ☐ **Q4 2025:** Mobile app with offline support
- ☐ **Q1 2026:** Advanced analytics dashboard

- ☐ **Q2 2026:** Integration with SAP SuccessFactors for enterprise events
- ☐ **Q3 2026:** Multi-language support with AI translation

## Contributing

This is a proprietary project. Please contact the repository administrators for contribution guidelines.

## License

This project is proprietary and not licensed for public use without explicit permission.

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Built with  for enhancing SAP event experiences

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